

Topics in Mathematics with Applications in Finance

MIT 18.S096 · In-Depth Study Guide
Full concept & formula explanations · Lectures 1–26

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Lecture 1 · Markets, Players & the Time Value of Money

The financial system is a machine for moving money across time and risk between people who have capital and people who need it. This lecture lays out who the players are, then builds the single most important quantitative idea in finance: a dollar today is worth more than a dollar tomorrow.

1.1 Markets & Products

A **market** is any venue (historically a local exchange, now mostly electronic) where buyers and sellers meet to trade. The things they trade — the **products** or asset classes — are:

- **Money / cash:** currencies (FX), and now digital money like BTC. The most liquid asset; the unit everything else is priced in.
- **Equities:** stocks (ownership shares in a company), ETFs (baskets of stocks), IPOs. The **primary market** is where new shares are first sold (pre-IPO / the IPO itself — money goes to the company); the **secondary market** is everyone trading those shares afterward (money goes between investors).

- **Credit / fixed income:** loans, bonds, mortgages. You lend money and get paid **interest**. The **interest rate** isn't a single number — it's a **curve** that depends on how long the money is held, and the whole curve moves over time.
- **Commodities:** physical goods — oil (energy), metals, agriculture.
- **Real estate & mortgages:** property and the loans secured against it.
- **Derivatives & structured products:** contracts whose value is *derived* from something else:
 - **Option** — the *right but not the obligation* to buy/sell an asset in the future at a set price.
 - **Future** — a standardized, exchange-traded obligation to buy/sell (often a commodity) at a set price and date.
 - **Forward** — like a future but private/customized (over-the-counter): agree now to buy an asset later.
 - **Swap** — exchange one stream of cash flows for another (e.g. fixed interest for floating).

1.2 Market Players

- **Commercial banks:** take deposits, make loans. The plumbing of the system.
- **Investment banks:** organized into **Equity** (stocks), **Fixed Income** (bonds, rates, FX, commodities — "FICC"), and **IBD / Corporate Finance** (helping companies raise capital, go public, issue bonds — a *services* business that earns fees, not trading profits).
- **Asset managers:** run mutual funds — invest other people's money against a benchmark. **Wealth managers** advise families/individuals.
- **Hedge funds:** trading entities that generate returns from their own + investors' capital, often with more freedom (leverage, shorting, derivatives).
- **Retail investors:** ordinary individuals.
- **Central banks:** guide markets & provide liquidity through policy (rates, QE). Not profit-seeking — they steer.
- **Corporates:** companies that trade mainly to **hedge** (e.g. an airline locking in fuel prices), not to speculate.

The core relationship: Investors *have* money and *need* returns. Borrowers *need* money and *have* risks/projects to fund. The trade between lender and borrower is the engine of the entire financial system. In fund structures, **LPs (Limited Partners)** are the investors who put up capital; **GPs (General Partners)** are the managers who run the strategy and take a cut.

1.3 Trader Types

- **Hedgers:** trade to *reduce* existing risk.
- **Market makers:** quote both a buy (bid) and sell (offer) price and earn the **bid-offer spread**. Their goal is to stay risk-neutral — earn the spread on flow, not bet on direction.
- **Proprietary traders & portfolio managers:** deliberately *take* risk with capital to seek return. This is the speculative end.

1.4 What Financial Mathematics Does

- **Pricing models:** derivatives are hard to price; we use differential equations and the principle of **relative value / arbitrage-free pricing** — a price is "right" if it doesn't let anyone make a risk-free profit.
- **Risk management:** position sizing, portfolio construction, capital allocation, leverage, liquidity, counterparty risk. The discipline is to *let the math speak* and remove emotion (greed/fear).
- **Trading strategies:** the dream is a "holy grail" — a perpetual money-making machine — implemented by a systematic "robo-trader." Reality is more humble.

Two big questions the course keeps circling: (1) Do people actually learn from experience, or just naively extrapolate historical patterns? (2) The **Efficient Market Hypothesis** vs **behavioral finance**: is trading a zero-sum game where all the "\$100 bills on the sidewalk" (easy profits in the data) have already been picked up? If markets are perfectly efficient, no strategy beats the index; in practice, frictions and behavior leave some bills behind.

1.5 Investment Strategy Choices

Every investor makes a series of either/or choices:

- **Internal (direct)** — invest yourself vs **External (fund-of-funds)** — hire someone else.
- **Public** assets (equity, bonds, FX, commodities — liquid, listed) vs **Private** assets (private equity, venture capital, buyouts, real estate, natural resources — illiquid).
- **Passive (Beta)** — just buy the benchmark/SPY and capture market return vs **Active (Alpha)** — pick securities to beat the market.
- **Systematic (quant)** — mathematical rules execute the trades vs **Discretionary** — a human decides.
- **Deterministic** — "if X happens then Y will happen" (rare and hard to find now) vs **Statistical** — "if X happens, Y is more *probable*."
- **Trend-following** — market goes up, buy more (momentum) vs **Mean-reversion** — market goes up, sell (bet it reverts).
- **Short-term** (fast, liquid) vs **Long-term** (slow, illiquid).

Value vs Growth: A **value** investor buys when current intrinsic value exceeds price, $V(0) > P$ — you're getting more than you pay. A **growth** investor bets future value will exceed today's, $V(T) > V(0)$ — you pay up today expecting the business to compound. (Notation: $V(0)$ = value now, $V(T)$ = value at future time T , P = price now.)

The meta-advice: understand your own objectives and constraints — *what you need the money for* and *when*. Then pick the area you're genuinely best at (your strength = your **edge**) and choose a strategy that fits it.

1.6 Time Value of Money — Compounding

This is the quantitative heart of Lecture 1. Money earns interest, so the *same* nominal amount is worth different things at different times. Start with \$1 and interest rate r per year:

After 1 year: $1 + r$ · After n years (annual compounding): $(1 + r)^n$

If instead interest is paid *twice* a year, each half-year you earn $r/2$, and over n years there are $2n$ periods:

$$\left(1 + \frac{r}{2}\right)^{2n}$$

Generalize to compounding m times per year:

$$\left(1 + \frac{r}{m}\right)^{mn}$$

Now let the compounding get infinitely frequent, $m \rightarrow \infty$. This limit is one of the most important in finance — it defines **continuous compounding**:

$$\lim_{m \rightarrow \infty} \left(1 + \frac{r}{m}\right)^{mn} = e^{rn}$$

Why it matters: e^{rn} is smooth and differentiable, which makes the calculus of continuous-time finance (Itô calculus, Black–Scholes) tractable. More compounding always grows money faster, but the gain saturates at the exponential e^{rn} — that's the ceiling.

1.7 Present Value & Discounting

Compounding pushes money *forward* in time. **Discounting** is the reverse: what is a future dollar worth *today*? Ask: if I'll receive \$1 in one year, how much (x) should I pay now? Fairness says borrowing x today and repaying it with interest should exactly equal the \$1 I receive:

$$x(1 + r) = 1 \implies x = \frac{1}{1 + r}$$

This x is the **present value** of \$1, and $\frac{1}{1+r}$ is the **discount factor**. Generalizing to a notional N received in n years:

$$\text{Discrete: } P = \frac{N}{(1 + r)^n} \quad \text{Continuous: } P = N e^{-rn}$$

Intuition: the discount factor is the "exchange rate between future and present dollars." A higher rate r or a longer horizon n makes future money worth less today. Every valuation in finance is ultimately a sum of discounted future cash flows.

1.8 Bonds

Zero-coupon (discount) bond

The simplest bond: the right to receive a fixed notional N at a single future date (the **maturity**), with no payments in between. Its price today is just the discounted notional, $P = N/(1 + r)^n$ — you buy it cheap and it "accretes" up to N .

Coupon bond

A normal bond pays periodic **coupons** C along the way and returns the notional N at maturity. From the issuer's view: borrow P now, pay coupons each period, repay N at the end. The price is the present value of *all* those cash flows. With annual coupons over n years:

$$P = C \frac{(1 + r)^n - 1}{r(1 + r)^n} + \frac{N}{(1 + r)^n}$$

The first term is a geometric series — the PV of the coupon stream (an **annuity**); the second is the PV of the notional repayment. The fraction $\frac{(1+r)^n - 1}{r(1+r)^n}$ is the annuity factor: it sums $\frac{1}{(1+r)}, \frac{1}{(1+r)^2}, \dots$ in closed form.

Yield

The **yield** y is the single discount rate that, plugged into the pricing formula, reproduces the bond's *current market price*. It's the bond's internal rate of return if held to maturity.

- For a bond trading at **par** (price = notional), the yield equals the coupon rate.
- For an n -year zero-coupon bond: $y = \frac{1}{\sqrt[n]{P}} - 1$ (with $N = 1$) — invert the discounting.
- **Bond price falls as yield rises, and rises as the coupon rises.** Price and yield move in opposite directions — this inverse relationship is fundamental.

Yield curve

Bonds of different maturities generally have different yields; plotting yield against maturity gives the **yield curve**. From the curve you can **bootstrap** discount factors for every horizon (strip out each cash flow's implied rate, shortest first). Normally the curve slopes *upward* — investors demand more yield to lock money up longer. When it **inverts** (long yields below short), it has historically been a reliable *predictor of recession*.

Price sensitivities: duration & convexity

How much does a bond's price move when yields move? Take derivatives of price w.r.t. yield.

Duration (first-order, % price change per unit yield): $D = -\frac{1}{P} \frac{dP}{dy}$

- Zero-coupon bond: $D_z = \frac{T}{1+y}$ — roughly its maturity. Longer bonds are more rate-sensitive.
- Continuously-compounded par bond: $D_p = \frac{1 - e^{-yT}}{y}$.

Convexity (second-order curvature): $C = \frac{1}{P} \frac{d^2 P}{dy^2}$

Intuition: duration is the *slope* of the price–yield curve (a linear approximation to how price responds), convexity is its *curvature*. Because the price–yield relationship is curved (convex), duration alone understates gains when yields fall and overstates losses when they rise; convexity corrects for that. The price move is approximately $\frac{\Delta P}{P} \approx -D \Delta y + \frac{1}{2} C (\Delta y)^2$. This is the bond-market version of a Taylor expansion — and the same delta/gamma idea reappears for options.

The lecture closes by pointing ahead to **Linear Algebra** — the mathematical language for handling many bonds, rates, and assets at once (Lecture 2).

Lecture 2 · Linear Algebra

Finance is full of many things at once — hundreds of stocks, dozens of rates. Linear algebra is the language for handling whole collections of numbers as single objects (vectors, matrices) and for finding hidden structure inside them (eigenvalues, PCA).

2.1 Vectors, Matrices, and What They Mean

A **vector** is an ordered list of numbers — e.g. the weights of each stock in your portfolio, or the returns of every asset on one day. A **vector space** is just the set of all such lists that you can add together and scale, and still stay inside the set. Key sub-ideas: a **basis** is a minimal set of directions that can build every other vector by combination; the number of them is the **dimension**; vectors are **linearly independent** if none is a combination of the others (no redundancy).

A **matrix** $A \in \mathbb{R}^{m \times n}$ is a grid of numbers, but the deeper meaning is that it's a **linear transformation**: feed in a vector x , get out Ax — a rotation, stretch, and/or projection of x . Two everyday operations:

- **Transpose** A^T flips rows and columns. A matrix is **symmetric** if $A = A^T$ — covariance matrices are always symmetric because $\text{Cov}(X, Y) = \text{Cov}(Y, X)$.
- **Inverse** A^{-1} undoes the transformation; it exists only when A doesn't collapse any dimension, i.e. when $\det A \neq 0$.

Trace $\text{tr}(A) = \sum_i a_{ii} = \sum_i \lambda_i$ **Determinant** $\det A = \prod_i \lambda_i$

The trace (sum of the diagonal) equals the sum of eigenvalues; the determinant (the volume-scaling factor of the transformation) equals their product. A zero determinant means the map squashes space flat — information is lost and it can't be inverted.

2.2 Eigenvalues & Eigenvectors

$$Av = \lambda v, \quad v \neq 0$$

An **eigenvector** v is a special direction that the matrix only *stretches* (by the scalar **eigenvalue** λ) without rotating. You find them by solving the **characteristic equation** $\det(A - \lambda I) = 0$, which gives the λ 's; each λ then yields its v .

Why finance cares: eigenvectors reveal the "natural axes" of a system. For a covariance matrix, the top eigenvector is the direction in which the portfolio of assets varies the most — the dominant risk factor. For a real *symmetric* matrix the eigenvalues are always real and the eigenvectors are orthogonal (perpendicular), which is exactly the clean structure we want.

2.3 The Spectral Theorem (the workhorse)

$$\text{Symmetric } A = Q\Lambda Q^T, \quad Q \text{ orthogonal } (Q^T Q = I), \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$$

Any symmetric matrix can be *diagonalized* by an orthogonal matrix: Q 's columns are the orthonormal eigenvectors, Λ holds the eigenvalues. Reading it right-to-left, A acts on a vector by (1) rotating into the eigenvector frame Q^T , (2) stretching each axis by its λ , (3) rotating back Q . The messy matrix is really just "stretch along the right axes."

Positive (semi)definite: A is **positive definite** if $x^T A x > 0$ for every nonzero x (positive *semidefinite* if ≥ 0). This holds exactly when all eigenvalues are positive (nonnegative). **Every valid covariance matrix is PSD** — because a variance, $x^T \Sigma x$, can never be negative.

$$\text{Portfolio variance as a quadratic form: } \sigma_p^2 = w^T \Sigma w \geq 0$$

Here w is the weight vector and Σ the covariance matrix. The whole risk of a portfolio collapses into this one quadratic form — that's why PSD matrices are everywhere in Lectures 7 and 14.

2.4 SVD & PCA

$$\text{SVD: any matrix } A = U\Sigma V^T \quad (U, V \text{ orthogonal; } \Sigma \text{ diagonal of singular values } \sigma_i \geq 0)$$

The **Singular Value Decomposition** generalizes the spectral theorem to *any* matrix (not just square/symmetric). It says every linear map is: rotate (V^T), scale by singular values (Σ), rotate again (U). The size of each σ_i tells you how important that direction is.

Principal Component Analysis (PCA) applies this to a covariance matrix: eigendecompose $\Sigma = Q\Lambda Q^T$ and keep only the top few eigenvectors (the **principal components**) — the directions holding most of the variance. This compresses many correlated variables into a few independent factors.

Classic finance result: run PCA on changes in the interest-rate yield curve and the first three components almost always come out as **level** (whole curve shifts up/down), **slope** (steepens/flattens), and **curvature** (the belly bends). Three numbers explain ~99% of all rate moves — a massive dimensionality reduction.

Cholesky decomposition $\Sigma = LL^T$ (with L lower-triangular) is the practical tool for *simulating* correlated random variables: draw independent standard normals $z \sim N(0, I)$ and form $X = Lz$; then X has exactly covariance Σ . This is the backbone of Monte-Carlo risk engines (Lecture 7).

Lecture 3 · Probability Theory

Markets are uncertain, so every model rests on probability. This lecture builds the vocabulary — random variables, expectation, variance, the key distributions — and ends with the two limit theorems that explain why the bell curve dominates finance.

3.1 The Foundations

Formally, randomness lives in a **probability space** $(\Omega, \mathcal{F}, P)$: Ω is the set of all possible outcomes (the **sample space**), $\mathcal{F}$ is the collection of **events** we can assign probabilities to (a σ -algebra — closed under complements and countable unions), and P is the **probability measure** assigning each event a number in $[0, 1]$. A **random variable** $X : \Omega \rightarrow \mathbb{R}$ is a function turning outcomes into numbers (e.g. tomorrow's return).

Its distribution is described by the **CDF** $F(x) = P(X \leq x)$ (probability of being at most x) and, when smooth, the **density** $f(x) = F'(x)$ (relative likelihood near x).

3.2 Expectation, Variance, Moments

$$\mathbb{E}[X] = \int x f(x) dx \quad \text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

The **expectation** $\mathbb{E}[X]$ is the probability-weighted average — the long-run mean, the center of mass of the distribution. **Variance** measures spread around that mean; its square root, the **standard deviation** σ , is what finance calls **volatility**. The handy identity "mean of the square minus square of the mean" comes straight from expanding $\mathbb{E}[(X - \mu)^2]$.

Moment generating function: $M_X(t) = \mathbb{E}[e^{tX}]$

The MGF is a single function whose derivatives at 0 spit out all the moments $(\mathbb{E}[X], \mathbb{E}[X^2], \dots)$, and it uniquely identifies the distribution — a compact fingerprint, useful for proving sums-of-variables results.

3.3 The Key Distributions

- **Bernoulli / Binomial:** a single yes/no trial, and the count of successes in n trials. (Up/down moves, defaults.)
- **Poisson(λ):** counts of rare events in a fixed interval at average rate λ — models jumps and defaults (Lecture 5).
- **Exponential:** waiting time between Poisson events; **memoryless** (the past doesn't change the expected wait).
- **Normal $N(\mu, \sigma^2)$:** the bell curve, $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$. Fully described by mean and variance, and *closed under linear combinations* (a sum of normals is normal) — which makes the math beautifully tractable.
- **Lognormal:** if $\log X$ is normal then X is lognormal. Stock prices are modeled this way because prices can't go negative and returns compound multiplicatively (Lecture 18).

3.4 Joint Behavior: Covariance, Correlation, Independence

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] \quad \rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \in [-1, 1]$$

Covariance measures whether two variables move together (positive) or oppositely (negative). **Correlation** ρ is covariance rescaled to a clean $[-1, 1]$, stripping out the units. These populate the covariance matrix Σ from Lecture 2.

Independence (one variable carries no information about the other) implies zero correlation — but the converse is *false* in general: variables can be uncorrelated yet dependent in nonlinear ways. The one big exception is the *jointly normal* case, where uncorrelated does mean independent.

Bayes' rule, $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$, tells you how to update beliefs when new information arrives.

3.5 The Two Limit Theorems

LLN: $\bar{X}_n \rightarrow \mu$ CLT: $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$

The **Law of Large Numbers** says sample averages converge to the true mean as you collect more data — the justification for estimating anything from history. The **Central Limit Theorem** is deeper: *regardless* of the underlying distribution, the sum (or average) of many independent pieces is approximately **normal**.

Why the bell curve rules finance: a price move is the aggregate of countless small independent influences, so the CLT predicts approximate normality — which is why so many models start there. The crucial caveat the course stresses: real returns have **fatter tails** than the normal (extreme moves happen far more often than the bell curve says), because real shocks aren't independent and finite-variance. That gap is the seed of volatility modeling (Lecture 9) and risk management (Lecture 7).

Lecture 4 · Matrix Primer

A practical follow-up to Lecture 2: the specific matrix types and decompositions you actually compute with in pricing and risk, and why numerical stability matters.

4.1 Special Matrices and Their Roles

- **Diagonal:** only diagonal entries — acts by independent scaling on each axis; trivial to invert.
- **Orthogonal** ($Q^T Q = I$): pure rotations/reflections, preserve lengths and angles. Their inverse is just the transpose — numerically perfect.
- **Idempotent** ($P^2 = P$): **projection** matrices — applying twice does nothing new because you're already on the subspace. The regression "hat matrix" is one.
- **Symmetric / positive-definite:** the covariance-matrix family from Lecture 2.

Projection (hat) matrix: $P = X(X^T X)^{-1} X^T$

This projects the data y onto the column space of X — geometrically, it finds the closest point reachable by a linear combination of your predictors. That projection is the fitted value in regression (Lecture 6); least squares is literally "drop a perpendicular."

4.2 The Decomposition Toolkit

LU (solve systems) · QR (least squares) · Cholesky $A = LL^T$ (PD only) · Eigen $A = Q\Lambda Q^T$ (symmetric) · SVD $A = U\Sigma V^T$ (any)

Each decomposition rewrites a matrix as a product of simpler ones so a hard problem becomes easy. **LU** turns solving $Ax = b$ into two triangular solves; **QR** stably solves least squares without forming $X^T X$; **Cholesky** is the fast "square root" of a PD matrix (used for simulation); **Eigen/SVD** expose the underlying axes and importances.

4.3 Quadratic Forms & Numerical Stability

The **quadratic form** $w^T \Sigma w$ (portfolio variance) is minimized subject to constraints in portfolio optimization (Lecture 14) — typically via Lagrange multipliers, which turn into a linear system.

Condition number: $\kappa = \frac{\sigma_{\max}}{\sigma_{\min}}$ (ratio of largest to smallest singular value)

Why this is a recurring villain: when assets are highly correlated, Σ has tiny eigenvalues, so κ is huge and the matrix is **ill-conditioned**. Inverting it amplifies noise enormously — tiny estimation errors in the inputs produce wild, unstable portfolio weights. This single problem motivates **regularization** (Lecture 10), shrinkage, and factor models (Lecture 15), all of which tame the inverse.

Lecture 5 · Stochastic Processes I

A stochastic process is randomness unfolding through time. This lecture introduces the four processes that underpin all of quantitative finance: Markov chains, martingales, the Poisson process, and Brownian motion.

5.1 What a Stochastic Process Is

A **stochastic process** $\{X_t\}$ is a family of random variables indexed by time — a "random function" whose value at each moment is uncertain. The **filtration** $\mathcal{F}_t$ formalizes "all information available up to time t "; it grows as time passes and we learn more. Everything we forecast is conditioned on $\mathcal{F}_t$.

5.2 Markov Chains

Markov property: $P(X_{n+1} \mid X_n, X_{n-1}, \dots, X_0) = P(X_{n+1} \mid X_n)$

A process is **Markov** if the future depends on the past *only through the present* — once you know today's state, the path that led there is irrelevant. This "memorylessness" makes models tractable. The dynamics are encoded in a **transition matrix** P , where P_{ij} is the probability of moving from state i to state j ; each row sums to 1.

Multi-step transitions are just matrix powers: the n -step probabilities are P^n (this is why Lecture 2's linear algebra matters). A **stationary distribution** π satisfies $\pi P = \pi$ — a long-run equilibrium the chain settles into, and it's exactly the left eigenvector of P for eigenvalue 1.

5.3 Martingales

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = X_n$$

A **martingale** is the mathematical "fair game": your best forecast of tomorrow's value, given everything you know, is simply today's value — no predictable drift up or down. A random walk with no edge is a martingale.

Why it's the keystone of pricing: the absence of arbitrage is *equivalent* to the existence of a probability measure (the "risk-neutral" measure, Lecture 20) under which discounted asset prices are martingales. If prices weren't a fair game under that measure, you could construct a money pump. Every derivative price in the course is ultimately a martingale expectation.

5.4 Poisson Process & Brownian Motion

These are the two fundamental continuous-time processes, one for *jumps* and one for *diffusion*.

- **Poisson process** (λ): counts discrete events (defaults, trades, jumps). The count by time t is $N(t) \sim \text{Poisson}(\lambda t)$; increments are independent; the waiting time between events is exponential. It moves in sudden unit jumps.
- **Brownian motion / Wiener process** W_t : the continuous-path counterpart. Defined by four properties:
 - $W_0 = 0$ (starts at the origin);
 - **independent increments** (non-overlapping moves are independent);
 - $W_t - W_s \sim N(0, t - s)$ (Gaussian increments whose variance grows with elapsed time);
 - continuous but **nowhere-differentiable** paths (infinitely jagged — you can't define an ordinary velocity).

The big payoff: Brownian motion is the random "engine" inside the stock-price model $dS = \mu S dt + \sigma S dW$. Because its paths are so jagged that ordinary calculus breaks ($(dW)^2$ doesn't vanish — it equals dt), we need a new calculus to handle it: Itô calculus (Lecture 18). Poisson processes add the jumps that pure Brownian motion misses.

Lecture 6 · Regression Analysis

Regression is how we fit relationships to data — estimating betas, hedge ratios, and factor loadings. It's the most-used statistical tool in finance, and it's pure linear algebra under the hood.

6.1 The Linear Model

$$y = X\beta + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2 I)$$

We model an outcome y (e.g. a stock's returns) as a linear combination of predictors — the columns of the design matrix X (e.g. market returns, factor returns) — weighted by unknown coefficients β , plus random noise ε . The noise is assumed to have zero mean, constant variance, and be uncorrelated across observations. The goal is to estimate β .

6.2 Ordinary Least Squares (OLS)

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

OLS chooses β to minimize the sum of squared residuals $\|y - X\beta\|^2$ — the total squared vertical distance between data and fit. Setting the derivative to zero gives the **normal equations** $X^T X \beta = X^T y$, whose solution is the formula above. Geometrically (tying back to Lecture 4), the fitted values $X\hat{\beta} = Py$ are the *projection* of y onto the space spanned by the predictors — the closest reachable approximation.

Gauss–Markov theorem: under the standard assumptions, OLS is **BLUE** — the Best (minimum-variance) Linear Unbiased Estimator. No other unbiased linear method beats it. And when the errors are normal, OLS coincides with maximum-likelihood estimation, so it's also the most probable explanation of the data.

6.3 Goodness of Fit & Inference

- $R^2 = 1 - \text{SSR}/\text{SST}$: the fraction of the variance in y explained by the model. $R^2 = 1$ is a perfect fit; 0 means the predictors explain nothing.
- t -**statistic** $= \hat{\beta}_j / \text{SE}(\hat{\beta}_j)$: tests whether a coefficient is real or could plausibly be zero (a large $|t|$, roughly > 2 , signals significance). The F -test checks the predictors jointly.

6.4 When the Assumptions Break

Real financial data routinely violate the textbook assumptions, and each violation has a name and a fix:

- **Heteroskedasticity**: error variance isn't constant (volatility clusters) → use robust standard errors.
- **Autocorrelation**: errors are correlated over time → Newey–West corrections.
- **Multicollinearity**: predictors are nearly redundant, making $X^T X$ ill-conditioned (Lecture 4's κ problem) → unstable, untrustworthy coefficients; fix with regularization (Lecture 10).
- **Endogeneity**: a predictor is correlated with the error, biasing the estimate.

Lecture 7 · Value at Risk (VaR)

VaR answers the question every risk manager and regulator asks: "How bad can it get on a normal bad day?" It distills a whole portfolio's risk into one dollar number — powerful, standard, and famously flawed.

7.1 The Definition

VaR_α is the loss level L such that $P(L > \text{VaR}_\alpha) = 1 - \alpha$

In words: with confidence α (say 99%), the loss over a chosen horizon (say one day) will *not* exceed the VaR. A "1-day 99% VaR of \$10M" means: on 99 of 100 days you lose less than \$10M; roughly one day in a hundred you lose more. It's a **quantile** of the loss distribution.

7.2 Three Ways to Compute It

- **Parametric (variance–covariance)**: assume returns are normal, so VaR is just a multiple of the standard deviation:

$$\text{VaR} = z_\alpha \sigma \sqrt{\Delta t} \cdot V$$

where z_α is the normal quantile ($z_{99\%} = 2.33$, $z_{95\%} = 1.65$), σ the volatility, and V the portfolio value. Fast and clean, but inherits the normal's thin tails — it *underestimates* extreme losses.

- **Historical simulation:** replay the actual returns of the last (say) 500 days on today's portfolio and read off the empirical quantile. Makes no distributional assumption and captures real fat tails — but assumes the past represents the future.
- **Monte Carlo:** simulate thousands of future scenarios for the risk factors (using Lecture 2's Cholesky to inject correlations), revalue the portfolio in each, and take the quantile. The most flexible (handles options' nonlinearity) but computationally heavy.

7.3 Beyond VaR: Expected Shortfall & Coherence

Expected Shortfall (CVaR): $ES_\alpha = \mathbb{E}[L \mid L > VaR_\alpha]$

VaR tells you a threshold but says *nothing* about how bad losses are once you cross it. **Expected Shortfall** fixes this by averaging the losses in the tail beyond VaR — it answers "if it's a bad day, how bad on average?"

Why VaR is "not coherent": a good risk measure should reward diversification — the risk of a combined book shouldn't exceed the sum of the parts (**subadditivity**). VaR can *violate* this: combining two portfolios can show *higher* VaR than the sum, wrongly penalizing diversification. Expected Shortfall is provably subadditive (coherent), which is why Basel regulation has shifted from VaR toward ES. Other limits of VaR: it's backward-looking, ignores tail shape, and is sensitive to model choice.

Lecture 8 · Time Series Analysis I

Time series methods model data that arrives in sequence, where today depends on yesterday. The core models — AR, MA, ARMA — are the foundation for forecasting returns, rates, and (next lecture) volatility.

8.1 Stationarity & Autocorrelation

Most time-series math requires **stationarity**: the statistical properties don't drift over time. **Weak (covariance) stationarity** needs three things — constant mean, constant variance, and an autocovariance $\text{Cov}(X_t, X_{t+h})$ that depends only on the lag h , not on absolute time. Without it, "yesterday" isn't comparable to "today" and estimates are meaningless.

$$\text{ACF: } \rho(h) = \frac{\gamma(h)}{\gamma(0)} = \frac{\text{Cov}(X_t, X_{t+h})}{\text{Var}(X_t)}$$

The **autocorrelation function** measures how strongly the series at lag h relates to itself — the memory in the data. **White noise** (pure randomness) has $\rho(h) = 0$ for all $h \neq 0$: no memory, unpredictable.

8.2 AR, MA, and ARMA Models

$$\text{AR}(p): X_t = \sum_{i=1}^p \phi_i X_{t-i} + \varepsilon_t \quad \text{MA}(q): X_t = \varepsilon_t + \sum_{j=1}^q \theta_j \varepsilon_{t-j}$$

An **autoregressive (AR)** model says today is a weighted sum of recent *values* plus a shock — the series remembers its own level. A **moving-average (MA)** model says today is a weighted sum of recent *shocks* — the memory is of past surprises. **ARMA(p,q)** combines both, and using the lag operator L (where $LX_t = X_{t-1}$) it writes compactly as $\phi(L)X_t = \theta(L)\varepsilon_t$.

The simplest case, **AR(1)**: $X_t = \phi X_{t-1} + \varepsilon_t$, is stationary only if $|\phi| < 1$ (otherwise shocks compound and the series explodes). Its autocorrelations decay geometrically, $\rho(h) = \phi^{|h|}$ — recent past matters most, distant past fades.

How you pick the model: the ACF and the **partial** ACF (PACF) act as fingerprints — the ACF cuts off sharply for an MA process, while the PACF cuts off for an AR process. You read these plots to choose p and q , then estimate the coefficients by Yule–Walker equations or maximum likelihood, and forecast by rolling the recursion forward. Forecast uncertainty grows with the horizon.

Lecture 9 · Volatility Modeling

Volatility isn't constant — it clusters into calm and stormy regimes. ARCH/GARCH models capture this, and they're central to risk, option pricing, and trading.

9.1 The Stylized Facts

Empirically, asset returns show a robust pattern: returns themselves are nearly *uncorrelated* (you can't predict direction), but they're *not independent* — **squared** returns are strongly autocorrelated. This is **volatility clustering**: big moves follow big moves, quiet follows quiet ("turbulent and calm regimes"). Returns also have **fat tails** — more extreme events than a normal allows.

9.2 ARCH and GARCH

$$\text{ARCH}(q): \sigma_t^2 = \alpha_0 + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2$$

The **ARCH** (AutoRegressive Conditional Heteroskedasticity) model makes today's *variance* a function of recent squared shocks: a big surprise yesterday raises today's expected volatility. This directly produces clustering.

$$\text{GARCH}(p,q): \sigma_t^2 = \alpha_0 + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2$$

GARCH generalizes ARCH by also letting today's variance depend on *past variances* — adding smooth persistence. The everyday workhorse is **GARCH(1,1)**: $\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$.

- **Persistence** = $\alpha_1 + \beta_1$; when it's close to 1, volatility shocks decay slowly (vol is "sticky").
- **Long-run variance** = $\frac{\alpha_0}{1 - \alpha_1 - \beta_1}$ — the level volatility mean-reverts toward.

- EWMA (RiskMetrics): $\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) \varepsilon_{t-1}^2$ — a GARCH special case with no long-run mean; simple and widely used.

Estimation & uses: GARCH parameters are fit by **maximum likelihood**. Refinements like **EGARCH** and **GJR-GARCH** add the **leverage effect** — volatility rises more after *down* moves than up moves (bad news scares markets more). Distinguish **realized volatility** (computed from high-frequency returns) from **implied volatility** (backed out of option prices, Lecture 19) — the gap between them is itself a tradable signal.

Lecture 10 · Regularized Pricing & Risk Models

When you fit models to noisy market quotes, naive least squares blows up. Regularization is the disciplined fix — trading a little bias for a lot of stability.

10.1 The Ill-Posed Problem

Calibrating a yield curve or a volatility surface to market prices means fitting many parameters to noisy, sometimes sparse quotes. As Lecture 4 warned, the matrix $X^T X$ is often near-singular (ill-conditioned), so the unconstrained solution swings wildly — small data changes produce huge parameter changes, and the model **overfits** noise instead of signal.

10.2 Ridge and Lasso

Ridge (L2): $\hat{\beta} = (X^T X + \lambda I)^{-1} X^T y$

Ridge regression adds λI to $X^T X$ before inverting. That $+\lambda I$ lifts all the eigenvalues away from zero, so the inverse is stable again — it directly cures the ill-conditioning. Equivalently, it penalizes large coefficients ($\lambda \|\beta\|_2^2$), shrinking them toward zero.

Lasso (L1): minimize $\|X\beta - y\|^2 + \lambda \|\beta\|_1$

Lasso penalizes the sum of *absolute* coefficients instead. The geometry of the L1 penalty drives many coefficients *exactly* to zero, so it performs automatic **feature selection** — handing you a sparse, interpretable model.

Tikhonov regularization generalizes ridge to penalize *roughness*, $\min \|X\beta - y\|^2 + \lambda \|\Gamma\beta\|^2$, forcing the fitted curve to be smooth — exactly what you want for an arbitrage-free yield or vol surface.

The bias-variance trade-off: the penalty strength λ is the dial. Small λ = flexible but noisy (high variance); large λ = stable but oversimplified (high bias). You choose λ by **cross-validation** — pick the value that predicts best on held-out data. In finance this delivers smooth surfaces, stable Greeks for hedging, and sensible bond/swap prices even when quotes are thin.

Lecture 11 · Time Series Analysis II

Extending Lecture 8 to multiple series at once, and to the powerful idea that two wandering series can be tethered together — the statistical basis of pairs trading.

11.1 Vector Autoregression (VAR)

$$\mathbf{X}_t = \mathbf{c} + \sum_{i=1}^p \Phi_i \mathbf{X}_{t-i} + \boldsymbol{\varepsilon}_t$$

A **VAR** models several time series jointly: each variable depends on the lags of *all* the variables (the Φ_i are now matrices). This captures feedback — e.g. how rates, inflation, and growth influence each other over time — and supports impulse-response analysis (how a shock to one variable propagates to the others).

11.2 Unit Roots & Cointegration

A **random walk** $X_t = X_{t-1} + \varepsilon_t$ has a **unit root** ($\phi = 1$) and is *non-stationary* — it wanders without reverting, and standard regression between two such series gives **spurious** (fake) relationships. The **Augmented Dickey-Fuller test** checks for a unit root.

Cointegration: two non-stationary series X_t, Y_t are cointegrated if some linear combination $\beta^T X_t$ is stationary.

Why traders love this: two stocks may each wander unpredictably (random walks), yet their *spread* stays mean-reverting and bounded — they share a long-run equilibrium. That's a **pairs trade**: when the spread stretches, bet it snaps back. The **error-correction model (ECM)**, $\Delta X_t = \alpha(\beta^T X_{t-1}) + \dots$, formalizes the pull back toward equilibrium, where α is the speed of correction. Engle-Granger and Johansen are the standard tests.

Lecture 12 · Time Series Analysis III

When the thing you care about is hidden — a true price, a latent volatility, a market regime — state-space models and the Kalman filter let you estimate it optimally from noisy observations.

12.1 State-Space Models

State (hidden): $x_t = Ax_{t-1} + w_t$

Observation (measured): $y_t = Cx_t + v_t$

A **state-space model** splits the world into a hidden **state** that evolves on its own (x_t) and noisy **observations** we actually see (y_t). The "true" fair value of an asset might be x_t , while traded prices y_t are that value plus microstructure noise v_t . The goal is to recover x_t from the y 's.

12.2 The Kalman Filter

The **Kalman filter** estimates the hidden state recursively in two repeating steps:

- **Predict:** use the state equation to project the current estimate and its uncertainty (covariance) one step forward.
- **Update:** when the new observation arrives, correct the prediction by a weighted amount — the **Kalman gain** K — that optimally balances trust in the model vs trust in the data (more measurement noise → less weight on the observation).

Why it's optimal & where it's used: for linear systems with Gaussian noise, the Kalman filter gives the provably minimum-mean-square-error estimate — it's the recursive Bayesian update done exactly. Finance uses: dynamic (time-varying) hedge ratios and betas, extracting latent factors, estimating stochastic volatility, and de-noising prices. Its relatives — **Hidden Markov Models** (with the EM algorithm) for regime-switching, and **particle filters** for nonlinear systems — extend the same idea.

Lecture 13 · Commodity Models

Commodities behave differently from stocks: they're physical, must be stored, and are consumed. That changes the relationship between spot and futures prices in important ways.

13.1 Spot, Futures & Cost of Carry

$$F = S e^{(r+u-y)T}$$

The futures price F links to the spot S through the **cost of carry**: financing cost r plus storage cost u minus the **convenience yield** y . The convenience yield is the intangible benefit of holding the physical commodity — having oil on hand to avoid a refinery shutdown is worth something a paper contract isn't. It behaves like a dividend.

- **Contango**: $F > S$ — futures above spot, an upward-sloping curve (normal when storage is plentiful and carry costs dominate).
- **Backwardation**: $F < S$ — futures below spot, typically when the commodity is scarce *now*, driving convenience yield high.

13.2 Spot Price Dynamics

$$\text{Schwartz mean-reversion: } d \ln S = \kappa(\theta - \ln S) dt + \sigma dW$$

Unlike stocks (which trend), commodity prices tend to **mean-revert** toward a long-run level θ tied to the marginal cost of production — when prices are high, producers pump more and prices fall, and vice versa. The parameter κ is the speed of reversion. Richer two- and three-factor models also let the convenience yield and interest rates be stochastic.

Extra features that matter: strong **seasonality** (natural gas in winter, agriculture at harvest), **storage constraints** that bound the curve, and price **spikes** in non-storable commodities like electricity (which can't be inventoried, so supply shocks pass straight to price). These break the clean stock-style models and demand commodity-specific structure.

Lecture 14 · Portfolio Theory

Markowitz's mean–variance framework launched modern finance: it shows mathematically how diversification works and what the "best" portfolio is. It leads directly to the CAPM.

14.1 Mean–Variance Optimization

$$\text{Return: } \mu_p = w^T \mu \quad \text{Risk: } \sigma_p^2 = w^T \Sigma w$$

A portfolio is a weight vector w . Its expected return is the weighted average of asset returns; its variance is the quadratic form $w^T \Sigma w$ from Lecture 2. The crucial insight lives in Σ 's off-diagonal terms: combining imperfectly-correlated assets makes portfolio variance *less* than the weighted average of individual variances — that's **diversification**, "the only free lunch in finance."

The optimization: *minimize variance for a given target return* (or maximize return for given risk), subject to $\sum w_i = 1$. Solving with Lagrange multipliers traces out the **efficient frontier** — a hyperbola in return–risk space whose upper edge holds the optimal portfolios (nothing offers more return at the same risk). The **two-fund theorem** says every efficient portfolio is a blend of just two fixed efficient portfolios.

14.2 Adding a Risk-Free Asset & the CAPM

Introduce a risk-free asset paying r_f . Now the best achievable combinations form a straight line — the **Capital Market Line** — drawn from r_f tangent to the efficient frontier. The tangency point is the **maximum-Sharpe portfolio**:

$$\text{Sharpe ratio: } \frac{\mu_p - r_f}{\sigma_p} \text{ — excess return per unit of risk.}$$

Everyone (in theory) holds the same tangency portfolio and just mixes it with cash to suit their risk appetite. If that tangency portfolio is the whole market, you arrive at the **Capital Asset Pricing Model**:

$$\mathbb{E}[R_i] = r_f + \beta_i (\mathbb{E}[R_m] - r_f), \quad \beta_i = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)}$$

The big lesson: only **systematic** risk (correlation with the market, measured by β) earns a return premium. **Idiosyncratic** risk is diversifiable, so the market doesn't pay you to bear it. An asset's fair expected return is the risk-free rate plus beta times the market's risk premium — nothing else. This is the conceptual seed of factor models (Lecture 15).

Lecture 15 · Factor Modeling

CAPM has one factor (the market). The real world has many. Factor models explain returns and risk through a handful of common drivers — and they tame the unstable covariance matrix.

15.1 The Structure

$$R_i = \alpha_i + \sum_k \beta_{ik} f_k + \varepsilon_i \quad \implies \quad \Sigma = B \Omega_f B^T + D$$

Each asset's return is its sensitivity (loading β_{ik}) to a set of common **factors** f_k , plus an idiosyncratic piece ε_i . The payoff is the covariance decomposition on the right: total risk = common factor risk ($B \Omega_f B^T$) + diversifiable idiosyncratic risk (D , diagonal).

Why this is a big deal numerically: estimating a full covariance matrix for 500 stocks needs $\sim 125,000$ parameters from noisy data — hopelessly ill-conditioned (Lecture 4). A 10-factor model needs only the loadings plus a small factor covariance — a few thousand stable parameters. Factor models are the practical cure for the unstable Σ that plagues portfolio optimization.

15.2 Types of Factors

- **Macroeconomic:** observable economic series — GDP growth, inflation, interest rates, oil. Intuitive but noisy.
- **Fundamental:** firm characteristics. The famous **Fama–French** model uses market, **size** (small minus big, SMB), and **value** (high minus low book/price, HML); later work adds **momentum** and quality. These are the workhorses of equity quant.

- **Statistical:** let PCA/factor analysis (Lecture 2) extract latent factors straight from the return data — maximally explanatory but hard to interpret.

$$\text{APT: no-arbitrage} \Rightarrow \mathbb{E}[R_i] = r_f + \sum_k \beta_{ik} \lambda_k$$

Arbitrage Pricing Theory generalizes CAPM: expected return is the risk-free rate plus exposure to each factor times that factor's risk premium λ_k . It's CAPM with many sources of priced risk.

Lecture 16 · Portfolio Management

From theory to practice: how active managers turn forecasts into portfolios, measure skill, and handle the noise that makes naive optimization unusable.

16.1 Measuring Active Skill

Alpha is return beyond the benchmark — the value an active manager adds. The right risk-adjusted measure is the **Information Ratio** $= \alpha/\omega$, active return divided by **tracking error** ω (the volatility of the active return). It's the Sharpe ratio of skill.

$$\text{Fundamental Law of Active Management: } IR \approx IC \cdot \sqrt{BR}$$

This elegant result decomposes skill into two parts: the **Information Coefficient** (IC) — how good your forecasts are (correlation between predictions and outcomes) — times the square root of **breadth** (BR) — the number of *independent* bets per year. The message: you can win either by being more accurate *or* by making more independent bets. A mediocre signal applied across thousands of stocks can beat a brilliant signal used a few times a year.

16.2 Building the Portfolio

Naive Markowitz optimization is notoriously unusable in practice — it amplifies estimation error in expected returns into extreme, unstable weights (the Lecture 4 ill-conditioning problem again). Fixes:

- **Black–Litterman:** start from the market-equilibrium portfolio (reverse-engineer the returns the market *implies*), then blend in your subjective *views* with confidence weights. The result is stable, intuitive, and only tilts away from the market where you have a real opinion.
- **Risk budgeting / risk parity:** allocate by each position's *risk contribution* rather than its dollar weight, so no single bet dominates.
- **Shrinkage** of Σ (Ledoit–Wolf), realistic **transaction-cost** and turnover penalties, and constraints (long-only, leverage caps) round out a usable process.

Lecture 17 · Stochastic Processes II

A deeper look at Brownian motion — the properties that make continuous-time finance work, especially the one strange fact $((dW)^2 = dt)$ that forces a whole new calculus.

17.1 Brownian Motion, Deeper

Brownian motion has remarkable structure. It's **self-similar**: $W_{ct} \stackrel{d}{=} \sqrt{c} W_t$ — zoom in or out and it looks statistically the same (a fractal). Several functions of it are **martingales** (fair games): W_t itself, $W_t^2 - t$, and the exponential $e^{\theta W_t - \theta^2 t/2}$. These martingales are the engines behind pricing proofs.

Quadratic variation: $[W, W]_t = t$, equivalently $(dW)^2 = dt$

The single most important fact in the course: for an ordinary smooth function, $(dx)^2$ is negligible compared to dx . But Brownian paths are so jagged that the sum of squared increments doesn't vanish — it accumulates to exactly the elapsed time t . So $(dW)^2$ behaves like dt , a first-order quantity. This is precisely why ordinary calculus fails for W_t and why Itô's lemma (Lecture 18) carries an extra second-derivative term.

17.2 Stopping Times & First Passage

- A **stopping time** τ is a random time whose "stop now" decision uses only information available so far (no peeking into the future) — e.g. the first time a price hits a barrier. The **optional stopping theorem** says a martingale stays "fair" even when stopped at such a time.
- The **reflection principle** cleverly computes the distribution of a Brownian path's running *maximum* and its **first-passage time** to a level — exactly what's needed to price **barrier options** and model **default times** (when a firm's value first hits zero).
- **Geometric Brownian motion** $S_t = S_0 e^{(\mu - \sigma^2/2)t + \sigma W_t}$ stays positive and has lognormal prices — the standard equity model.

Lecture 18 · Itô Calculus

The calculus of random paths. Itô's lemma is the chain rule for functions of Brownian motion, and it's the tool that produces the Black–Scholes equation.

18.1 The Itô Integral

We can't integrate against Brownian motion the ordinary way (its paths aren't differentiable). The **Itô integral** $\int_0^t H_s dW_s$ is defined by evaluating the integrand H at the *left* endpoint of each tiny interval — the **non-anticipating** choice, meaning you decide your position *before* seeing the next random move (exactly like trading: you size your position, then the market moves). A beautiful consequence: the Itô integral is itself a **martingale**, so $\mathbb{E}[\int H dW] = 0$. The **Itô isometry** $\mathbb{E}[(\int H dW)^2] = \mathbb{E}[\int H^2 ds]$ lets us compute its variance.

18.2 Itô's Lemma — the Chain Rule for Randomness

For $dX = \mu dt + \sigma dW$ and a function $f(t, X)$:

$$df = \left(f_t + \mu f_x + \frac{1}{2} \sigma^2 f_{xx} \right) dt + \sigma f_x dW$$

This is the centerpiece. Compared to the ordinary chain rule, the *extra* term $\frac{1}{2} \sigma^2 f_{xx}$ appears because, by Lecture 17, $(dW)^2 = dt$ doesn't vanish — a second-order Taylor term survives into first order. Intuitively, a convex function ($f_{xx} > 0$) benefits from volatility (the "wiggling" of X pushes its average up), and Itô's lemma quantifies exactly that boost. This term is the mathematical home of **gamma** and option convexity.

Worked application — the stock model. Take $f = \ln S$ with geometric Brownian motion $dS = \mu S dt + \sigma S dW$. Here $f_x = 1/S$, $f_{xx} = -1/S^2$. Plugging in:

$$d \ln S = \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dW$$

Integrating gives $S_t = S_0 e^{(\mu - \sigma^2/2)t + \sigma W_t}$ — the lognormal price. Note the $-\frac{1}{2}\sigma^2$ **drift correction**: it's why the average *log* return is below the average simple return — a direct, famous consequence of Itô. The product rule also changes: $d(XY) = X dY + Y dX + dX dY$, with a new cross-term $dX dY$.

Lecture 19 · Black–Scholes Formula

The crown jewel: a closed-form option price derived from a no-arbitrage hedging argument. It earned a Nobel Prize and built the modern derivatives industry.

19.1 The Hedging Argument & the PDE

The key idea is **replication**. Hold the option and short Δ shares to form a portfolio $\Pi = V - \Delta S$. Choosing $\Delta = \partial V / \partial S$ cancels the random dW term (via Itô's lemma) — the portfolio becomes *instantaneously riskless* over the next instant. By no-arbitrage, a riskless portfolio must earn the risk-free rate r . Setting its return to r and rearranging gives the **Black–Scholes PDE**:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

Remarkably, the stock's real drift μ has *vanished* — the option's price doesn't depend on whether you think the stock will rise, only on its volatility. That's the seed of risk-neutral pricing.

19.2 The Formula for a European Call

$$C = S_0 N(d_1) - Ke^{-rT} N(d_2)$$
$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

Solving the PDE with the call's payoff boundary gives this. Reading it: $S_0 N(d_1)$ is the expected value of receiving the stock if the option finishes in-the-money; $Ke^{-rT} N(d_2)$ is the discounted strike times the **risk-neutral probability** $N(d_2)$ of exercise. $N(\cdot)$ is the normal CDF. **Put–call parity** $C - P = S_0 - Ke^{-rT}$ then gives the put for free.

19.3 The Greeks

- **Delta** $\Delta = N(d_1) = \partial V / \partial S$: sensitivity to the stock — the hedge ratio (how many shares to hold).
- **Gamma** $\Gamma = \partial^2 V / \partial S^2$: how fast delta changes — the convexity, and the Itô term made tangible. High gamma = frequent re-hedging.
- **Theta** $\Theta = \partial V / \partial t$: time decay — options lose value as expiry nears.
- **Vega** $= \partial V / \partial \sigma$: sensitivity to volatility — the only unobservable input, so traders quote options in **implied volatility**.
- **Rho** $= \partial V / \partial r$: sensitivity to interest rates.

Risk-neutral valuation: because hedging removed μ , we can price as if all assets drift at r and discount at r : $V = e^{-rT} \mathbb{E}^Q[\text{payoff}]$. This "risk-neutral" trick (Lecture 20) generalizes far beyond Black–Scholes and is the universal pricing recipe.

Lecture 20 · Option Price & Probability Duality

A profound two-way street: option prices and probability distributions encode the same information. Prices tell you the market's implied probabilities, and vice versa.

20.1 Pricing as Expectation

$$V_0 = e^{-rT} \mathbb{E}^Q[\text{payoff}(S_T)]$$

Generalizing Lecture 19: every derivative's price is the discounted *expected* payoff — but the expectation is taken under a special **risk-neutral measure** Q , not real-world probabilities P . Under Q , all assets drift at the risk-free rate and discounted prices are martingales (Lecture 5).

The Fundamental Theorems of Asset Pricing: (1) **No arbitrage** $\Leftrightarrow$ there exists an equivalent martingale measure Q . (2) The market is **complete** (every payoff can be replicated) $\Leftrightarrow$ that Q is *unique*. This is the rigorous foundation under all derivative pricing: prices come from a martingale measure, whose existence is the no-arbitrage condition.

20.2 Recovering the Distribution (the Duality)

$$\text{Breedon-Litzenberger: } q(K) = e^{rT} \frac{\partial^2 C}{\partial K^2}$$

This is the stunning result. The second derivative of the call price with respect to *strike* K recovers the entire **risk-neutral probability density** of the future price S_T . So if you observe option prices across all strikes, you can read off the market's *complete* implied probability distribution — no model needed.

Real option markets show a **volatility smile/skew** (implied vol varies by strike), which by this duality means the implied distribution is *not* lognormal — it has fatter tails and negative skew (the market prices in crash risk that Black–Scholes ignores). **Dupire's local volatility** $\sigma_{loc}(K, T)$ is the unique volatility function that reprices the whole observed surface, used for calibration and exotic pricing.

Lecture 21 · Stochastic Differential Equations

SDEs are the differential equations of random systems — the general language for modeling how prices, rates, and volatilities evolve, with a toolkit (Feynman–Kac) linking them back to PDEs.

21.1 The General SDE

$$dX_t = \mu(X_t, t) dt + \sigma(X_t, t) dW_t$$

An SDE has two parts: a **drift** μ (the deterministic trend, the "where it's pulled") and a **diffusion** σ (the random noise, the "how much it shakes"). Letting μ and σ depend on the current state X and time produces a rich family of models. Under mild conditions a unique solution exists.

21.2 The Essential Models

- **Ornstein–Uhlenbeck (mean reversion):** $dX = \theta(\mu - X) dt + \sigma dW$. The drift pulls X back toward its long-run mean μ at speed θ — perfect for interest rates, spreads, and volatility, which don't wander off forever. It's Gaussian and stationary. (Vasicek's rate model is exactly this.)
- **Cox–Ingersoll–Ross (CIR):** $dX = \theta(\mu - X) dt + \sigma\sqrt{X} dW$. The $\sqrt{X}$ makes volatility shrink as $X \rightarrow 0$, keeping it *positive* — essential for rates and variance, which can't go negative.

21.3 The Bridge to PDEs

Feynman–Kac: the expectation $u(x, t) = \mathbb{E}[\text{payoff}(X_T) \mid X_t = x]$ solves a parabolic PDE.

This is the deep two-way link: solving an expectation over random paths is *equivalent* to solving a deterministic PDE. Black–Scholes is the famous special case — you can price either by simulating paths (Monte Carlo) or by solving the PDE on a grid; Feynman–Kac guarantees they agree. The **Kolmogorov forward (Fokker–Planck)** equation evolves the probability density itself, while the backward equation evolves expectations. Numerically, the **Euler–Maruyama** scheme $X_{t+\Delta} = X_t + \mu\Delta + \sigma\sqrt{\Delta} Z$ simulates SDE paths (Milstein adds higher-order accuracy).

Lecture 22 · Calculus of Variations & FX Execution

How do you optimize over an entire path or schedule, not just a number? The calculus of variations does exactly that — and its killer application is optimal trade execution.

22.1 Calculus of Variations

Minimize $J[y] = \int_a^b L(t, y, y') dt$ over functions y .

Euler–Lagrange:
$$\frac{\partial L}{\partial y} - \frac{d}{dt} \frac{\partial L}{\partial y'} = 0$$

Ordinary calculus minimizes over *numbers*; the calculus of variations minimizes over *functions* — finding the whole curve $y(t)$ that makes an integral (a **functional**) as small as possible. The **Euler–Lagrange equation** is the "set the derivative to zero" condition for this infinite-dimensional problem; any optimal path must satisfy it.

22.2 Optimal Execution (Almgren–Chriss)

The trading problem: you must liquidate a large position over a day. Trade too *fast* and your own selling pushes the price against you (**market impact** — costly). Trade too *slow* and you're exposed to price moves while you wait (**timing/volatility risk** — uncertain). The optimal **trading trajectory** $x(t)$ balances the two.

Minimize $\mathbb{E}[\text{cost}] + \lambda \text{Var}[\text{cost}]$ over the schedule $x(t)$

The solution & its meaning: Euler–Lagrange yields an **exponentially-decaying** liquidation schedule for a risk-averse trader (front-load to cut risk), collapsing toward a uniform **TWAP** (trade evenly through time) as risk-aversion $\lambda \rightarrow 0$. The parameter λ is the trader's risk

appetite. The same variational machinery governs **FX execution** scheduling — the lecture's named application — and underlies modern algorithmic execution desks.

Lecture 23 · Quanto Credit Hedging

A specialist topic showing how correlation between markets — here, foreign exchange and credit default — creates subtle risks that require careful, measure-theoretic hedging.

23.1 Quanto Products

A **quanto** is a derivative whose underlying is in one currency but whose payoff is delivered in another at a *fixed* exchange rate. For example, a contract on a euro-denominated asset that pays out in dollars at a pre-set FX rate. The buyer gets the asset's performance without the FX risk — but that risk doesn't vanish; it transfers to the issuer as a **correlation** exposure between the asset and the FX rate.

23.2 Change of Measure & the Quanto Adjustment

Girsanov / quanto drift adjustment: $-\rho \sigma_S \sigma_{FX}$

Pricing a quanto requires changing the **numeraire** (the currency you measure value in), and **Girsanov's theorem** tells you that switching measures *shifts the drift* of your processes. The specific shift is the **quanto adjustment** $-\rho \sigma_S \sigma_{FX}$ — proportional to the correlation ρ between the asset and FX, times both volatilities. If they're uncorrelated, no adjustment; if correlated, the expected growth rate under the pricing measure moves.

The credit application — wrong-way risk: sovereign **credit default swaps** are often quoted in a foreign currency (e.g. protection on Italy quoted in dollars). When a country defaults, its currency typically *collapses at the same moment* — default and FX jump together (a strong, dangerous correlation). This **quanto CDS basis** directly encodes the market's expected FX devaluation upon default. Hedging combines CDS with FX forwards/options and actively manages the **cross-gamma** (the correlation exposure) — you're hedging a risk that only shows up when two bad things happen together.

Lecture 24 · HJM Model for Interest Rates & Credit

Earlier rate models pin down one number (the short rate). The Heath–Jarrow–Morton framework models the entire forward-rate curve at once, with a remarkable no-arbitrage constraint.

24.1 Modeling the Whole Curve

Instead of modeling just the short rate and deriving the curve, **HJM** directly specifies the dynamics of every **forward rate** $f(t, T)$ — the rate, seen today at time t , for instantaneous borrowing at a future date T . This models the full term structure's evolution simultaneously, which is what you need to price interest-rate derivatives consistently.

$$df(t, T) = \alpha(t, T) dt + \sigma(t, T) dW_t$$

24.2 The No-Arbitrage Drift Condition

$$\text{HJM drift condition: } \alpha(t, T) = \sigma(t, T) \int_t^T \sigma(t, u) du$$

The key insight: you do *not* get to choose the drift α freely. No-arbitrage *forces* the drift to be completely determined by the **volatility structure** σ . This is the rate-curve analogue of "drift becomes r under the risk-neutral measure" — once you specify how volatile the forward rates are, the entire arbitrage-free drift is pinned down. So an HJM model is really just a choice of volatility function.

24.3 Connections & Credit Extension

The instantaneous **short rate** is just $r_t = f(t, t)$. Classic short-rate models are special cases of HJM with particular volatility choices: **Ho–Lee** (constant vol), **Hull–White** / **Vasicek** (mean-reverting), **CIR** (square-root). The same framework extends to **credit**: model forward **default/hazard rates** $\lambda(t, T)$ the same way, with the survival probability given by $\exp(-\int_t^T \lambda du)$ — a unified "defaultable HJM" for pricing risky bonds and credit derivatives.

Lecture 25 · Ross Recovery Theorem

A beautiful and controversial result: can we extract the market's real-world beliefs and risk preferences from option prices alone? Ross says yes — under strong assumptions, using the Perron–Frobenius theorem.

25.1 The Question

Market prices reveal the **risk-neutral** measure Q , but Q is a tangle of two things: people's *actual* probability beliefs P and their *risk preferences* (the pricing kernel). Normally you can't separate them from prices alone — you'd need to assume a utility function. Ross's bold claim: under the right structure, prices alone suffice to **recover** the real-world P .

25.2 The Recovery

$$\text{State prices: } P_{ij} = \delta \frac{\phi_j}{\phi_i} F_{ij}$$

Here F_{ij} are the real-world transition probabilities (what we want), δ is the discount factor, and ϕ is the **pricing kernel** (marginal utility of consumption in each state). Ross's key assumption is **transition independence** — the kernel depends only on the states, not the path. Under it, the matrix of observed state prices can be uniquely factored.

The Perron–Frobenius mechanism: the theorem (a linear-algebra result, Lecture 2) guarantees that a positive matrix has a *unique* positive eigenvalue with a positive eigenvector. Applying it to the state-price matrix extracts a unique solution for the discount δ , the kernel ϕ , and hence the real-world probabilities P — all from option prices. The payoff: implied real-world expected returns and risk premia, read straight off the options market.

The controversy: recovery requires a finite, recurrent Markov state and the strong transition-independence assumption. Borovička, Hansen & Scheinkman argued these assumptions effectively assume away a martingale component of the pricing kernel, so the "recovered" measure isn't the true physical one in general. A gorgeous theorem whose real-world validity is hotly debated.

Lecture 26 · Counterparty Credit Risk

The risk that the party on the other side of your derivative defaults while owing you money. The 2008 crisis made this central, and it's now priced into every trade as CVA.

26.1 The Nature of the Risk

When you hold an over-the-counter derivative, its value swings positive and negative over time. If it's *positive* (the counterparty owes you) and they **default**, you lose that value. This is **counterparty credit risk** — distinct from market risk because it depends jointly on market moves *and* the counterparty's creditworthiness, and it's bilateral (both sides bear it).

26.2 Exposure & CVA

Exposure: $E(t) = \max(V(t), 0)$ Expected Exposure: $EE(t) = \mathbb{E}^Q[E(t)]$

You only lose when the contract is in your favor, so **exposure** is the *positive* part of the value — like being long an option on the counterparty's default. **Expected Exposure** averages this over future scenarios. The **Potential Future Exposure (PFE)** is a high quantile of it — VaR applied to exposure.

$$\text{CVA} = (1 - R) \int_0^T EE(t) dPD(t)$$

The **Credit Valuation Adjustment** is the market price of this default risk — a charge subtracted from the trade's value. It integrates the expected exposure against the counterparty's **probability of default** PD (from their CDS spreads), scaled by the loss given default $(1 - R)$ where R is the recovery rate. Its mirror images: **DVA** (the value of your own default risk) and **FVA** (funding cost) — together the "XVA" family.

Wrong-way risk & mitigants: the dangerous case is **wrong-way risk** — when exposure *rises* precisely as the counterparty's credit *deteriorates* (the two are positively correlated, as in the quanto-CDS example of Lecture 23). It amplifies losses just when you can least afford them. The defenses: **netting sets** (offset positive and negative trades with the same counterparty), **collateral / CSA** agreements (post margin as exposure grows), **central clearing**, and default funds — the post-2008 architecture for containing systemic counterparty risk.

Study Map — How It All Connects

- **The mathematical spine:** Linear Algebra (2, 4) + Probability (3) provide the language. Everything statistical — Regression (6), Time Series (8, 11, 12), Volatility (9), Regularization (10) — is built on them. The recurring villain is the *ill-conditioned covariance matrix*, which links Lectures 4, 10, and 15.
- **Risk & portfolios:** Probability → VaR/ES (7) for measuring risk; Portfolio Theory (14) → Factor Models (15) → Portfolio Management (16) for building portfolios. CAPM is the one-factor seed of the multi-factor world.
- **Continuous-time pricing — the course's climax:** Stochastic Processes (5, 17) establish Brownian motion and the crucial $(dW)^2 = dt$; Itô Calculus (18) gives the chain rule; Black–Scholes (19) applies it to price options; Duality (20) reveals prices = probabilities; SDEs (21) generalize the dynamics. Risk-neutral pricing (martingales, Lecture 5) threads through all of them.
- **Advanced & applied:** Commodities (13), Optimal Execution (22), Quanto Credit (23), HJM rates/credit (24), Ross Recovery (25), and Counterparty Risk (26) each apply the core machinery to a specialized, real-world desk problem.

These notes are for understanding and review — pair them with the OCW lecture videos and problem sets for the full derivations and worked numerical examples. Concepts deliberately echo across lectures; noticing those echoes (e.g. convexity in bonds → gamma in options → the Itô term) is the fastest path to mastery.